TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
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Simplex Method Project

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The Simplex Algorithm

The Simplex Algorithm is a mathematical method used to solve linear programming problems. It was developed by George Dantzig in 1947 and has since become one of the most widely used optimization techniques in various fields, including operations research, economics, and engineering.

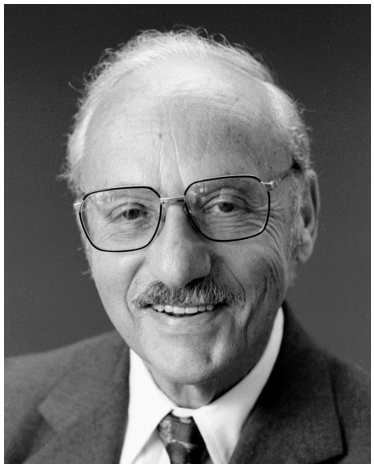


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

Original Problem

Description

- Problem Name: dsfsdfdsfsd
- Amount of decision variables: 3
- Amount of restrictions: 3
- Total Amount of variables: 6
- Type of problem: Maximization

Equations

$$
\begin{aligned}
Max\ Z &= 5.000000x_0 + 5.000000x_0 \\
such\ that: & \\
& 3.000000x_1 + 4.000000x_2 \geq 4.000000 \\
& 4.000000x_1 + 3.000000x_2 \geq 5.000000
\end{aligned}
$$

Initial Table

Z	x_0	x_0	s_1	s_2	b
1.0000	-5.0000	5.0000	0.0000	0.0000	0.0000
0.0000	3.0000	4.0000	1.0000	0.0000	4.0000
0.0000	4.0000	3.0000	0.0000	1.0000	5.0000

Final Table

Z	x_0	x_0	s_1	s_2	b
1.0000	0.0000	11.6667	1.6667	0.0000	6.6667
0.0000	1.0000	1.3333	0.3333	0.0000	1.3333
0.0000	0.0000	-2.3333	-1.3333	1.0000	-0.3333

Solution

Solution

The optimal solution is:

- $x_0 = 1.3333$
- $x_0 = -0.3333$

The optimal value of the objective function is: $Z = 6.6667$